302, 1302 (21|11|13)

BTS – III – 11.13 – 1163

B.Tech. Degree III Semester Examination November 2013

IT/ME/EC/EB/EI 302 ELECTRICAL TECHNOLOGY

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) Draw and explain the phasor diagram of practical transformer when its is connected to a capacitive load.

(b) Derive the emf equation of a DC generator.

(c) Explain critical field resistance and critical speed from open circuit characteristics of DC shunt generator.

(d) Explain cross magnetising and demagnetizing effect of armature reaction in DC generators.

(e) Discuss on pitch factor, pole pitch distribution factor and coil span with respect to an Alternator.

(f) A 6 pole induction motor is fed from 50Hz supply. If the frequency of rotor emf at full load is 2Hz find full load speed and slip.

(g) Derive the condition for maximum starting torque for 3 phase induction motor.

(h) Explain classification of substations.

PART B

(4 x 15 = 60)

II. (a) Derive the condition for maximum efficiency of single phase transformer. (3)

(b) A 15 KVA, 2300/230V, 50Hz single phase transformer gave the following test data (12)
OC – 2300V, 0.21A, 50IN
SC – 47V, 6A, 160W
 (i) Find the equivalent circuit referred to high voltage side
 (ii) Calculate full load voltage regulation at 0.8pf lagging when the load voltage is held at 230V.

OR

III. (a) Explain the working of auto transformer with diagram. (5)

(b) A 230/460V transformer has a primary resistance of 0.2Ω and a reactance of 0.5Ω and corresponding values for the secondary are 0.75Ω and 1.8Ω respectively. Find the secondary terminal voltge when supplying (i) 10A at 0.8 pf lagging (ii) 10A at .8pf leading. (10)

IV. (a) Explain power flow diagram of DC generator. (5)

(b) A short shunt compound generator supplies a load current of 100A at 250V. The generator has the following winding resistances shunt field 130Ω, armature 0.1Ω and series field 0.1Ω. Find the emf generated, if brush drop is 1V per brush. (10)

OR

V. (a) Discuss on various methods of speed control of DC series motors. (5)

(b) A 250V shunt motor runs at 100rpm at no load and taken 8A. The total armature and field resistances are 0.2Ω and 250Ω respectively. Calculate the speed when loaded and taking 50A. Assume flux to be constant. (10)

(P.T.O.)

VI. (a) Explain the working of synchronous motor at leading and lagging loads. (5)

(b) A 3 phase star connected 1000KVA, 11000V alternator has 52.5A. The AC resistance of winding per phase is 0.45Ω. The test results are given below: (10)
OC – field current = 12.5A, voltage between lines = 422V
SC – field current = 12.5A, Line current = 52.5A.
Determine the full load voltage regulation of alternator at (i) .8pf lag (ii) .8pf lead

OR

VII. (a) Explain the classification of 3 phase AC motors. (5)

(b) A 440V, 50HZ 3 phase induction motor draws an input power of 76KW from the mains. The rotor emf takes 120 complete cycles/minute. Its stator losses are 1KW and rotor current per phase is 62A. Calculate (i) rotor copper losses per phase (ii) torque developed (iii) rotor resistance per phase. (10)

VIII. (a) Explain the working of Thermal Power Plant with neat schematic diagram. (12)

(b) Explain different types of insulators in power system. (3)

OR

IX. (a) Explain different DC transmission schemes. (5)

(b) Explain economic load dispatch. (5)

(c) Discuss on various switch gears in power system. (5)

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